Component Diagram - Submit/Resubmit Document

«container»
Web Application
[React]

Runs in the browser. Shows only the sections each role can use: My Documents, Review Queue, All Documents, and Submit Document, plus the search bar, the review screen with the AI feedback, the profile, and in-system notifications. For the Super Admin: Audit Trail, User Accounts, Holiday Calendar, and AI Settings.

Sends submit, resubmit, and cancel requests
[HTTPS / JSON]

Backend Application [Laravel / PHP (Apache application server)]
[container]

«component»
Access Middleware
[Laravel Middleware]

Checks the active session and ends it after 60 minutes of no activity or when the account is disabled. Allows only the sections, actions, and documents permitted by the role, ownership, or reviewer assignment.

Passes allowed requests (Document Source, L1, and L2 only)

«component»
Submission Controller
[Laravel Controller]

Handles Submit/Resubmit Document and Cancel Submission. Provides the document types and the active Immediate Supervisors (L1) without the submitter, validates the details, attachments, change note, and cancel reason, allows a cancel only before the L1 opens the document or while it is Returned to Source, and returns the return details and the confirmation.

Checks the account's role and status
[SQL]

Checks for duplicates

Queues the AI feedback request

Saves the document or revision and assigns the L1, or cancels the submission

«component»
Duplicate Check Service
[Laravel Service]

Creates the hash of every attachment and compares it with existing documents, or with the last revision on resubmission. Saves a duplicate warning without blocking the submission.

«component»
AI Feedback Job
[Laravel Queued Job]

Runs Provide AI Feedback & Suggestions in the background. Skips it when the AI Assistant is off or the document type is not selected (never for Other). Extracts the text of the files or the linked document, adds the matching Active AI knowledge, and saves the feedback for each review level.

«component»
Workflow & Routing Service
[Laravel Service]

Generates reference numbers, saves documents and revisions, assigns reviewers, and changes the status following the L1, L2, L3 review order. Also cancels submissions, and moves waiting documents to a replacement reviewer or the new Division Chief (L3), returning them only if no one can take them.

Checks sharing with the L1; reads the content and last modified time

Saves the uploaded files

Notifies the assigned L1 (not on cancel)

Records submissions, resubmissions, and cancellations with the reason

Reads linked document text

Reads uploaded file text

Starts or stops the L1's TAT

«component»
Google Workspace Client
[Laravel Socialite / Google API Client]

Connects to Google Workspace for Sign in with Google, link sharing checks, and reading linked documents and their last modified time. Keeps the Google Drive permission of the document roles encrypted.

«component»
Document File Service
[Laravel Service]

Saves uploaded files to private storage, and opens or downloads them only after an access check.

«component»
Notification Service
[Laravel Service / Queued Job]

Saves the in-system notification and sends the matching email in the background through Gmail, recording any delivery failure. Sends only the email to an account that was disabled.

«component»
Audit Logging Service
[Laravel Service]

Writes read-only audit records with a server-generated timestamp, including the old and new values when something is changed.

«component»
TAT Service
[Laravel Service]

Starts and stops each reviewer's TAT and counts it in working days (Monday to Friday), not counting the holidays in the Holiday Calendar or the time waiting for the submitter to revise.

Sends text and AI knowledge; receives feedback per level
[HTTPS / REST API]

Compares hashes; saves warnings
[SQL]

Reads AI settings and knowledge; saves AI feedback
[SQL]

Writes documents, revisions, assignments, and status
[SQL]

Checks sharing; reads linked documents
[OAuth 2.0 / Google Drive API]

Sends emails through Gmail
[SMTP]

Saves and reads uploaded files
[Laravel Filesystem]

Saves in-system notifications
[SQL]

Writes audit records
[SQL]

Saves TAT dates; reads holidays
[SQL]

«external_system»
AI Assistant

AI model accessed through an API that gives preliminary feedback and suggestions for each review level.

«external_system»
Google Workspace

Handles Sign in with Google, checks the sharing of Google Workspace document links, and delivers email notifications through Gmail.

«container»
Uploaded File Storage
[Private Laravel file storage (application server disk)]

Keeps uploaded PDF, DOCX, and XLSX files (up to 10MB each and 100MB per submission) exactly as submitted. Files are opened or downloaded only through the backend after an access check.

«container»
DocuFlow Database
[MySQL (database server)]

Stores user accounts and roles with the encrypted Google Drive permission, documents and their revisions, Google Workspace links, file details and locations, document hashes, review records and remarks, AI feedback, AI knowledge and settings, the holiday calendar, in-system notifications, the audit trail, and the background job queue.